SRUN: Spectral Regularized Unsupervised Networks for Hyperspectral Target Detection

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***Abstract*— The high dimensionality of a hyperspectral image (HSI) provides the possibility of deeply capturing the underlying and intrinsic characteristics in spectra, such that targets embedded in the background can be detected. However, redundant information, deteriorated bands, and other interferences from background challenge the target detection problem. In this article, an effective feature extraction method based on unsupervised networks is proposed to mine intrinsic properties underlying HSIs. Our approach, called spectral regularized unsupervised networks (SRUN), imposes spectral regularization on autoencoder (AE) and variational AE (VAE) to emphasize spectral consistency, which is more suitable for characterizing spectral information of HSIs by hidden nodes than the original AE and VAE models. Then, we conduct a simple feature selection algorithm on the hidden nodes in the deepest code to select specific nodes that contain distinguishability between target and background, which is based on the spectral angular difference between a known target spectrum and spectra of other pixels in input. The selected nodes are further weighted adaptively to obtain a discriminative map depending on the observation that each selected node provides different contribution rates to target detection. Experimental results on several data sets illustrate that the proposed SRUN-based target detection algorithm is suitable for targets at the subpixel level and those with structural information.**

***Index Terms*— Background suppression, hyperspectral target detection, spectral regularization, unsupervised feature learning, variational autoencoders (VAEs).**

Manuscript received July 18, 2019; revised August 19, 2019 and October 8, 2019; accepted October 9, 2019. Date of publication October 29, 2019; date of current version January 21, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61801359, Grant 61571345, Grant 91538101, Grant 61501346, Grant 61502367, and Grant 61701360, in part by the Young Talent Fund from the University Association for Science and Technology, Shaanxi, China, under Grant 20190103, in part by the Special Financial Grant from the China Postdoctoral Science Foundation under Grant 2019T120878, in part by the 111 Project under Grant B08038, in part by the Fundamental Research Funds for the Central Universities under Grant JB180104, in part by the Natural Science Basic Research Plan in Shaanxi Province, China, under Grant 2019JQ153, Grant 2016JQ6023, and Grant 2016JQ6018, in part by the General Financial Grant from the China Postdoctoral Science Foundation under Grant 2017M620440, in part by the Yangtse Rive Scholar Bonus Schemes under Grant CJT160102, and in part by the Ten Thousand Talent Program. *(Corresponding authors: Weiying Xie; Gang He.)*

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Digital Object Identifier 10.1109/TGRS.2019.2947033

I. Introduction

HYPERSPECTRAL images (HSIs) can describe the detailed spectrum reflected by different classes or objects in a scene with hundreds of contiguous spectral channels. Benefitting from such rich spectral information, HSIs exhibit apparent advantages for a wide array of applications in military and civilian fields [1]–[4], where target detection plays a crucial role. Hyperspectral target detection is essentially a binary classification problem where pixels are identified as either target or background according to its unique spectral characteristic [5].

Many algorithms have been developed for target detection in HSI, such as matched filter (MF) [6], adaptive coherent/cosine estimator (ACE) [7], orthogonal subspace projection (OSP) [8], constrained energy minimization (CEM) [9], hierarchical CEM (hCEM) [10], and a combined sparse and collaborative representation (CSCR) based [11]. The MF and ACE algorithms consider target detection as hypothesis test problems. The OSP method is representative of the subspace-based methods. The CEM and its variation, hCEM, impose a constraint on the targets or backgrounds and build a finite-impulse response filter, which minimizes the output energy of the filter. The CSCR method is a typical sparse representation-based target detection method. Recently, multiple-instance (MI) spectral MF and MI-ACE were presented to estimate target signatures in HSIs from imprecisely labeled training data [12].

In reality, directly utilizing a vast amount of spectral information is difficult to meet the requirement of target detection owing to the similarity of spectra between target pixels and background pixels in wide-scale remote sensing imagery. For example, the pixels of aircraft in an airport scene captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor have similar spectra with background pixels, as shown in Fig. 1, while the spectra of different pixels from the aircraft are different. The reason for this phenomenon is that the targets have rare presence, and their spectra are susceptible to the influence of surrounding pixels, especially targets at the boundary. The small number of target pixels makes its accurate statistical estimation difficult.

As a consequence, two common practices are generally taken. One is feature selection [13], which selects an optimal subset of hyperspectral bands while discarding all unselected bands. The other is feature extraction [14], which can transform a high-dimensional data into a lower dimensional space

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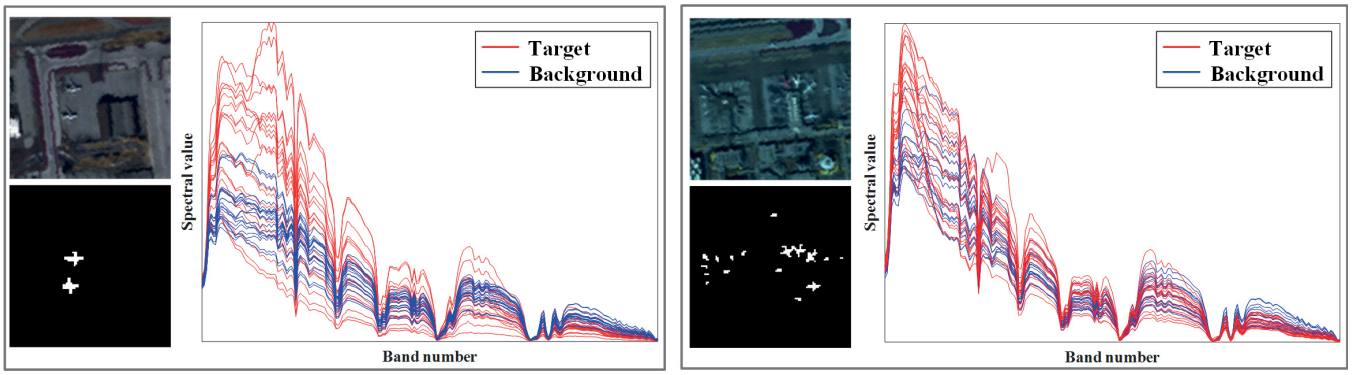


Fig. 1. Illustration of spatial and spectral information of several pixels in two airport scenes. The spectra of the aircraft targets are demonstrated in red line, and the background pixels are in blue line.

and learn the relatively discriminative features. Both of these two practices can reduce redundancy or noisy information, and thus, can improve the detection performance to a certain degree. In general, feature selection has a theoretic limitation. To be specific, an optimal subset for detection cannot be determined until all possible subset combinations have been carried out, which is an infeasible task because the subset combination of features to be examined increases exponentially with the number of dimensions [14]. Hence, this article mainly focuses on feature extraction with respect to nonlinear HSIs.

Two types of feature extraction approaches, including unsupervised approaches, such as principal component analysis and independent component analysis, and supervised approaches, such as linear discriminant analysis, are widely used [15]–[18]. These feature extraction methods extract shallow features in a linear way. Due to the limited representation capability of shallow and linear features, this category is not suitable for analyzing rare target samples in complex scenes. Recently, deep neural networks (DNNs) have shown an explosive success in high-level feature extraction to well characterize nonlinear HSIs. The extracted features have made significant contributions to the subsequent applications, such as classification [19]–[23], and anomaly detection [24]–[27]. The supervised DNNs, including convolutional neural network and generative adversarial network (GAN) [28], are widely exploited to extract effective features for classification [29]. Different from classification, rare labeled samples can be used for target or anomaly detection. The unsupervised DNNs, such as autoencoder (AE) [30], the variants of AE [31]–[33], and deep belief network (DBN) [34], [35], can learn useful representations for unlabeled data that are of increasing interest to improve detection performance. Ma *et al.* [25] considered the reconstruction errors between the input and the output of DBN caused by anomaly score directly. Zhao and Zhang [26] exploited AE to extract feature matrices of HSIs for anomaly detection.

Recently, deep generative models are the most consistent with the goal of implementing more effective feature representation to reveal the underlying structure of data. On account of the limited samples for HSIs, unsupervised leaning will become the future trend. AE-based unsupervised learning methods have yielded state-of-the-art methods for approximating the input, which is the key to solve the problem of target detection in HSIs because of the limited samples and time-consuming caused by high-dimensional spectral vectors. In particular, variational AEs (VAEs) [31] not only generate approximated data to the input but also provide a compact data representation.

In this article, AE and VAE models are developed and first applied to excavate the latent information in HSI for further target detection. We first build two spectral regularized unsupervised networks (SRUN) based on AE and VAE. Compared with the original AE and VAE, these two networks are more suitable for characterizing the spectral information of HSIs. There is no guarantee that all the hidden nodes extracted by the proposed SRUN contain distinguishability between target and background without any prior information. Therefore, we present a simple and effective feature selection method based on spectral angular difference between a known target spectrum and spectra of other pixels to select nodes that may contain relatively rich information of targets. Hence, the nodes containing target information can be effectively distinguished from all the other nodes. Based on the observation that each selected node provides a different contribution to target detection, the selected nodes are further weighted adaptively to obtain a discriminative map. In order to further decrease false alarm rate, the morphological opening operation is conducted on the weighted map to retain targets while suppressing background. Experimental results demonstrate the outstanding performance of the proposed SRUN-based target detection method in terms of detection accuracy and false-positive rate. Furthermore, the SRUN based on VAE can achieve superior performance than the SRUN based on AE.

In summary, the main technical contributions of this article are as follows.

1) We build up SRUN, a novel unsupervised feature extraction scheme specifically for HSIs that introduces spectral regularization into the loss function of the original AE and VAE models. It is, to the best of our knowledge, the first time the spectral consistency of such unsupervised networks during parameter updating for target detection is explored. We show that the proposed SRUN model

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is more suited for spectral feature extraction compared with the original AE and VAE models.
2) Given a target spectrum, we present a simple and powerful feature selection method to identify several specific nodes in a high-level layer that may contain relatively rich information of targets, which are further weighted adaptively to obtain a discriminative and comprehensive map.
3) The proposed SRUN-based target detection algorithm is suitable for targets presented at the pixel level and those with structural information. Experimental results on several data sets show that our framework achieves state-of-the-art performance compared with other competing methods.

In the rest of this article, Section II reviews the AE and VAE models. Section III introduces the proposed SRUN-based target detection method. Section IV shows the superior performance of the proposed method to other competing methods, and Section V draws the conclusion of this article.

II. Related Work

A. Autoencoders

An AE is a typical feed-forward neural network containing encoder and decoder in which the parameters (weight matrix and bias) are estimated by backpropagation [36]. The AE and its variations have adopted into HSI classification [37] and anomaly detection [38]. In an AE, the encoder aims at mapping the visible layer or input layer $\mathbf{x}$ into a hidden layer $\mathbf{z}$ with a weight matrix $\mathbf{W}$ and bias $\mathbf{b}$

$$\mathbf{z} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b}) \tag{1}$$

where σ represents an activation function. The decoder is used to map the hidden layer $\mathbf{z}$ to the reconstructed layer $\hat{\mathbf{x}}$ with a weight matrix $\mathbf{W}^d$ and bias $\mathbf{b}^d$

$$\hat{\mathbf{x}} = \sigma(\mathbf{W}^d\mathbf{z} + \mathbf{b}^d). \tag{2}$$

The AE encodes structure in the input data by the hidden layer so that it can be correctly decoded in the output. The parameters are obtained by minimizing the following cost function:

$$L_{\mathrm{AE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2. \tag{3}$$

The weight matrix and bias are updated by the mini-batch stochastic gradient descent [39] algorithm. By symmetrically stacking more layers, a hierarchical representation of input data can be learned. This is a stacked AE, which is considered as a deep architecture.

B. Variational Autoencoders

The VAE and its improved version have recently shown an explosively popularity for its success in unsupervised detection of lesions [40] and anomaly detection in videos [41]. However, as far as we know, there is no work for hyperspectral target detection based on VAE and its improved version. Here, we will briefly describe the principle of VAE. For a complete description of VAE and its applications, we refer readers to [31].

Different from GAN, VAE can approximate the posterior distribution in the latent space with a probabilistic encoder network $q_\phi(\mathbf{x}|\mathbf{z})$ and model the likelihood of input with a decoder network $p_\theta(\mathbf{z}|\mathbf{x})$. In a VAE, the loss function is the combination of the reconstruction error and the KL-divergence regularization term between the latent space vector and the representation with a mean vector and a standard deviation vector. The KL-divergence regularization term aims at optimizing the variational lower bound on the marginal log-likelihood of each observation. To generalize, the objective of VAE can be written as

$$L_{\mathrm{VAE}} = \sum_{i=1}^{N} -D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}^i)\|p_\theta(\mathbf{z})) + \frac{1}{L}\sum_{l=1}^{L}(\log p_\theta(\mathbf{x}^i|\mathbf{z}^{i,l})) \tag{4}$$

where $\mathbf{X}=\{\mathbf{x}^i\}_{i=1}^N$ represents the input data, $\mathbf{z}$ is the latent feature, and L is the number of auxiliary noise variables. The first term on the right-hand side of (4) is the KL-divergence regularization term and the second one is the reconstruction error in the traditional AE model. In the case of both $p_\theta(\mathbf{z})$ and $q_\phi(\mathbf{z}|\mathbf{x})$ are Gaussian, the resulting loss function can be calculated as

$$L_{\mathrm{VAE}} \simeq \frac{1}{2}\sum_{j=1}^{J}\left(1 + \log\left(\left(\sigma_j^i\right)^2\right) - \left(\mu_j^i\right)^2 - \left(\sigma_j^i\right)^2\right) + \frac{1}{L}\sum_{l=1}^{L}(\log p_\theta(\mathbf{x}^i|\mu_{\hat{\mathbf{x}}^{i,l}}, \sigma_{\hat{\mathbf{x}}^{i,l}})) \tag{5}$$

where $\mathbf{z}^{i,l}=\mu^i+\sigma^i\odot\varepsilon^l$ and $\varepsilon^l \sim \mathrm{N}(0, \mathbf{I})$. Here, the symbol "$\odot$" means elementwise product and $\hat{\mathbf{x}}$ represents the reconstructed version of $\mathbf{x}$. In a VAE model, the input spectral vector $\mathbf{x}^i$ passes through the encoder and generates the mean μ^i and variance σ^i of the $\mathbf{x}^i$ distribution, and the latent feature vector $\mathbf{z}^{i,l}$ is sampled based on the formula $\mathbf{z}^{i,l}=\mu^i+\sigma^i\odot\varepsilon^l$. Then, $\mathbf{z}^{i,l}$ is passed into the decoder to reconstruct output $\hat{\mathbf{x}}^i$ as same as $\mathbf{x}^i$ distribution.

III. Proposed Method

Fig. 2 shows the schematic of the proposed method, which consists of the following steps. First, we propose the SRUN to extract compact features in HSIs. Then, the effective nodes are selected and further weighted adaptively. Finally, the background is suppressed to obtain the detection map. More details of each step are described in the following.

A. Feature Extraction by SRUN

Lots of extensions based on the AE or VAE model have been investigated to enable better reconstruction quality and interpretable latent feature representation. For HSIs, we identify a relevant limitation of such unsupervised networks, lack of consistency in spectral vector. In this article, we develop SRUN based on AE and VAE for the compact representation of the high-dimensional HSIs during parameters updating that characterizes the key spectral information in a nonlinear way.

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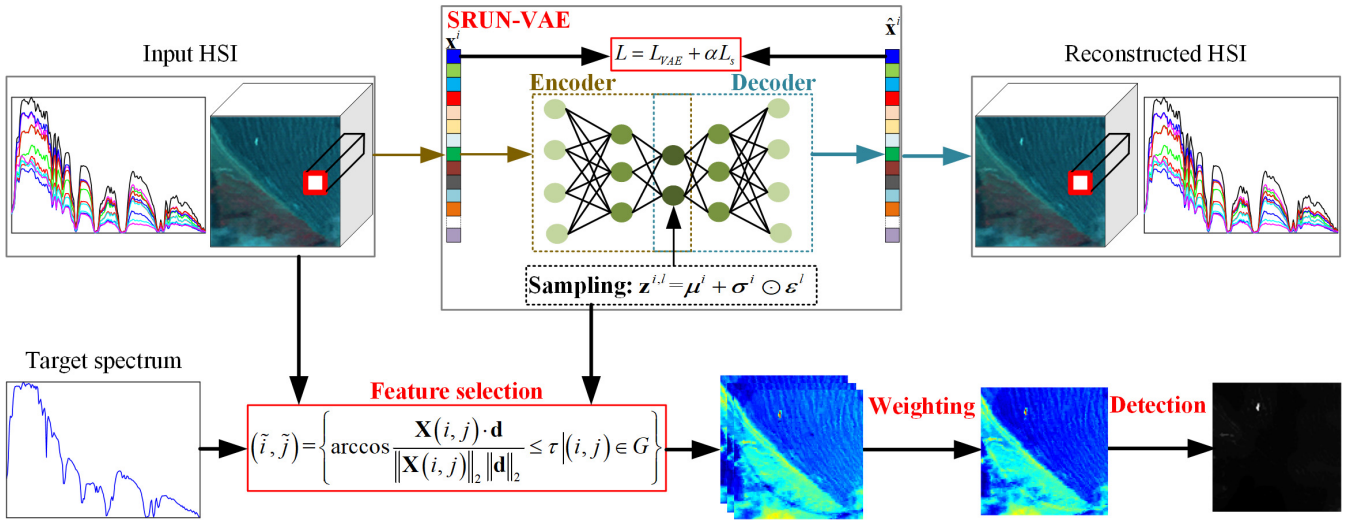


Fig. 2. Schematic of the proposed SRUN-based target detection method.

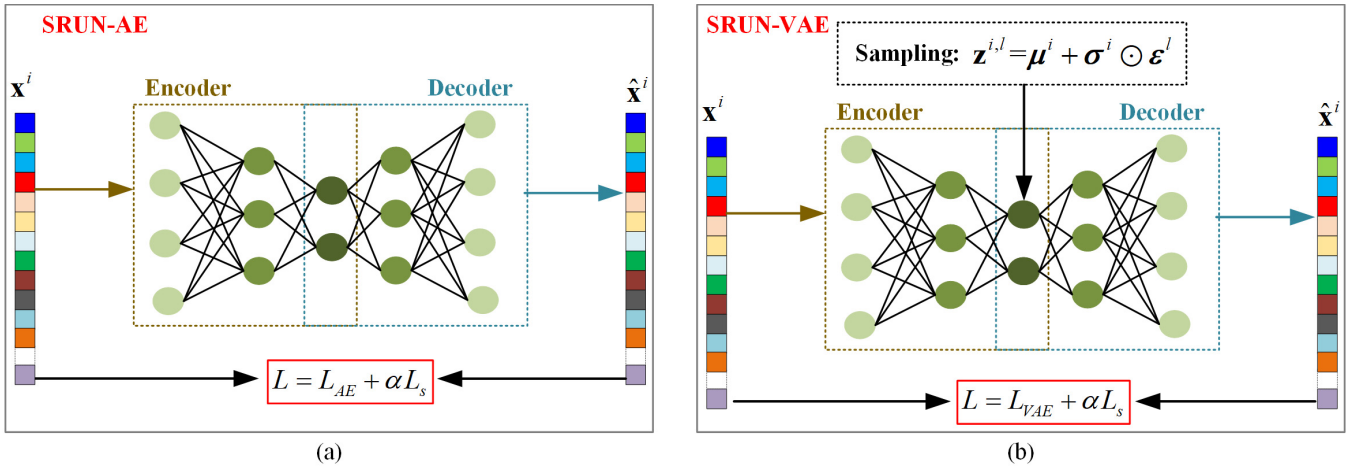


Fig. 3. Details of the proposed SRUN based on (a) AE, i.e., SRUN-AE and (b) VAE, i.e., SRUN-VAE. The SRUN minimizes two terms to update the parameters—the original objective of AE or VAE and the additional spectral regularization.

Besides encouraging the representation layer to follow L_{AE} and L_{VAE}, we further reduce the angular difference of the representations of the input and its reconstruction version during the optimization procedure. Thus, the proposed SRUN can be formulated as the following optimization problem:

$$L = L_{\mathrm{AE}} \vee L_{\mathrm{VAE}} + \alpha L_s \tag{6}$$

where L_s is the introduced spectral regularization term, α is a regularization parameter penalizing large L_s and "$\vee$" means OR operation. Fig. 3(a) and (b) shows details of the proposed SRUN based on AE and VAE, respectively. The spectral regularization term is identified by the spectral similarity between the input and reconstructed vectors, which can be defined as

$$L_s(\mathbf{x}^i, \hat{\mathbf{x}}^i) = \frac{1}{\pi} \arccos \frac{\mathbf{x}_i \cdot \hat{\mathbf{x}}_i}{\|\mathbf{x}_i\|_2 \|\hat{\mathbf{x}}_i\|_2}. \tag{7}$$

It is apparent that the spectral regularization term is the spectral angle between $\mathbf{x}^i$ and $\hat{\mathbf{x}}^i$, which can penalize the spectral difference between $\mathbf{x}^i$ and $\hat{\mathbf{x}}^i$. It is worth noting that the spectral angle is divided by the circular constant π to map the range of the angle from 0 to 1. The lower the L_s, the more similar the two vectors. In this way, the extracted features can represent the key spectral information to a large extent.

B. Feature Selection and Weighting

Fig. 4 shows the latent feature maps extracted by the proposed SRUN-AE and SRUN-VAE models, where 20 feature maps corresponding to 20 hidden nodes in the high-level of one testing image are displayed. It can be observed from Fig. 4 that different hidden nodes provide different levels of information of interest. In several hidden nodes, the information of targets can present more distinctive patterns as compared with that of the background. Besides, targets in HSIs are rare, and not a substantial number of features contain notable information of targets. The target feature maps (highlighted with yellow in Fig. 4) behave differently from those of the background.

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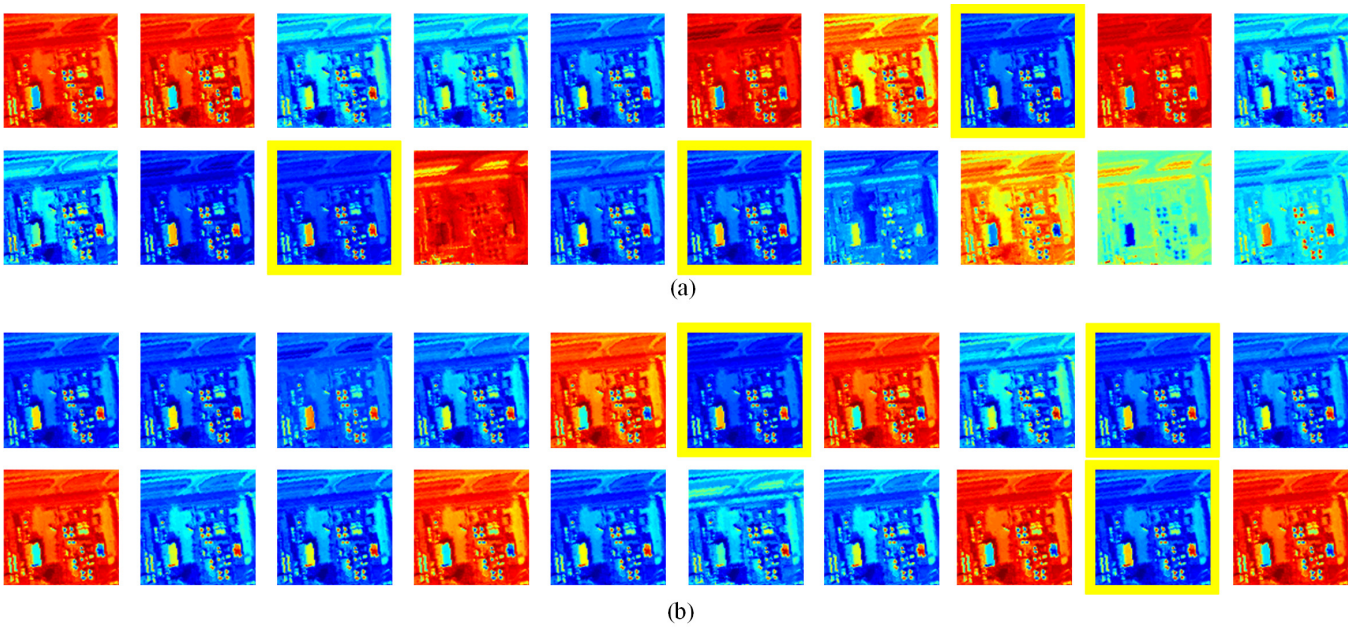


Fig. 4. Intermediate results of hidden nodes in visual by the (a) SRUN-AE and (b) SRUN-VAE with two layers. It is noted that there are 20 hidden nodes in the second hidden layer. The selected feature maps are highlighted with yellow.

Therefore, the primary objective of feature selection is to select an appropriate node subset from all the hidden nodes to well represent the information of interest. The information preserved by feature selection has a significant impact on target detection because the information of unselected nodes will be completely discarded after feature selection. Therefore, a key success in feature selection is how to design effective criteria for feature selection to meet target detection.

Given a target spectrum, we present a novel method to identify target feature maps in high-level layer that contain distinguishability between target and background. Let $\mathbf{Z}=[\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_C]$ represent the extracted feature map of an HSI data matrix $\mathbf{X}$. The proposed feature selection method satisfied the following constraint:

$$(\tilde{i}, \tilde{j})=\left\{\arccos \frac{\mathbf{X}(i, j) \cdot \mathbf{d}}{\|\mathbf{X}(i, j)\|_2\|\mathbf{d}\|_2} \leq \tau \,|\, (i, j) \in G\right\} \tag{8}$$

where $\mathbf{d}$ is the prior-known target spectrum, τ is a threshold, and G records the locations of maximum values in each $\mathbf{z}_c$ $(c = 1, 2, ..., C)$. When the spectral angle between $\mathbf{d}$ and $\mathbf{X}(i, j)$ is less than or equal to τ, the location of such spectral vectors are identified. Besides, the identified location comes from the set G that is the location of the maximum value in $\mathbf{z}_c$. Therefore, the feature map in $\mathbf{Z}$ corresponding to location $(\tilde{i}, \tilde{j})$ can be selected. If the maximum value in $\mathbf{z}_c$ is the target pixel, $\mathbf{z}_c$ will be selected. Conversely, $\mathbf{z}_c$ will not be selected. It is worth noting that we set a threshold to select several feature maps rather than one feature map by the minimum spectral angle between $\mathbf{d}$ and $\mathbf{X}(i, j)$; in this way, it is sufficient to detect targets with different sizes.

The selected feature maps are highlighted in Fig. 4. It can be observed that the feature maps corresponding to targets demonstrate two characteristics. One is that the target pixels often present around a small area. The other one is that the target pixels often display high contrast to the background pixels. These patterns can be effectively utilized to further detect targets. To combine the complementary information of the selected feature maps, we utilize an adaptive weighting method based on structure tensor

$$\hat{\mathbf{z}}=\sum_{k=1}^{K} \rho_k \mathbf{z}_k \tag{9}$$

where $[\rho_1, \rho_2, \ldots, \rho_K]^{\mathrm{T}}$ is the weight coefficient for each selected feature map $\mathbf{z}_k$. The structure tensor of the kth feature map at pixel i can be expressed as

$$\mathbf{T}_i^k = \begin{bmatrix} \left(\mathbf{Z}_i^k\right)_x^2 & \left(\mathbf{Z}_i^k\right)_x\left(\mathbf{Z}_i^k\right)_y \\ \left(\mathbf{Z}_i^k\right)_x\left(\mathbf{Z}_i^k\right)_y & \left(\mathbf{Z}_i^k\right)_y^2 \end{bmatrix} \tag{10}$$

where $\mathbf{T}_i^k$ is the structure tensor of the kth feature map at pixel i, and $(\mathbf{Z}_i^k)_x=(\partial \mathbf{Z}^k/\partial x)$ and $(\mathbf{Z}_i^k)_y=(\partial \mathbf{Z}^k/\partial y)$ represent the derivatives of the kth feature map at pixel i along x- and y-directions, respectively. This structure tensor $\mathbf{T}_i^k$ of the kth feature map can be decomposed as

$$\mathbf{T}_i^k = \begin{bmatrix} \eta_{1,i}^k & \eta_{2,i}^k \end{bmatrix} \begin{bmatrix} \lambda_{1,i}^k & 0 \\ 0 & \lambda_{2,i}^k \end{bmatrix} \begin{bmatrix} \eta_{1,i}^k \\ \eta_{2,i}^k \end{bmatrix} \tag{11}$$

where $\lambda_{1,i}^k$ and $\lambda_{2,i}^k$ are the nonnegative eigenvalues of kth feature map at pixel i, and $\eta_{1,i}^k$, $\eta_{2,i}^k$ are the corresponding eigenvectors of kth feature map at pixel i. Let $\lambda_{1,i}^k$ be the relative larger eigenvalue of kth feature map. As we know, the larger eigenvalue illustrates the edge intensity of kth feature map, and the corresponding eigenvector indicates the orientation that maximize the gray-value fluctuations. If $\lambda_{1,i}^k > \lambda_{1,i}^{k+1}$, the edge intensity of the kth feature map is larger at pixel i, that is,

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the kth feature map contains more information. Therefore, it should occupy a larger proportion of weight coefficient. Conversely, the $(k+1)$th feature map should occupy a larger proportion of weighting in the case of $\lambda_{1,i}^{k} < \lambda_{1,i}^{k+1}$. When the eigenvalues of all the selected feature maps are equal, all the feature maps have the same weight coefficient at this pixel, i.e., $1/K$. It can be obtained from the above-mentioned analysis that the weight coefficient for each selected feature map at pixel i can be calculated by

$$\rho_{k,i} = \frac{\lambda_{1,i}^{k}}{\sum_{k=1}^{K} \lambda_{1,i}^{k}}, \quad k = 1, 2, \ldots, K \tag{12}$$

where $\rho_{k,i}$ represents the weight of the kth feature map. Thus, the weight coefficient of each selected feature map can be gained by (12) and a comprehensive feature map can be adaptively weighted by (9). For a complete description of the weight coefficient, please refer to our previous article [42], [43].

C. Background Suppression

This section aims at suppressing background in the weighted map that contains discriminative patterns of targets. As mentioned in Section III-B, the feature map corresponding to targets contains two characteristics, i.e., rare presence and high contrast in comparison to background. Considering that the morphological opening operation can remove the bright (high contrast) connected components with specific area, we utilize such operation to suppress background in the weighted map

$$D = |\hat{\mathbf{z}} - O(\hat{\mathbf{z}})| \tag{13}$$

where $O(\hat{\mathbf{z}})$ represents the morphological opening operation on $\hat{\mathbf{z}}$. The term $|\hat{\mathbf{z}} - O(\hat{\mathbf{z}})|$ preserves the high contrast connected components in small area.

To further refine the background suppression map, we utilize a guided filter to take the local smooth prior to consideration [42]. The final detection result is given by

$$D_f = \bar{a}_i D_i + \bar{b}_i \tag{14}$$

where ($\bar{a}$ and $\bar{b}$) are the average values of linear coefficients in different local windows. The linear coefficients can be solved by minimizing the squared difference between the input and the filtering output in a local window. The filter size and blur degree of guided filter are set to 3 and 0.5 for all test data sets, respectively, learned from experience.

IV. Experimental Results and Analysis

A. Data Sets Description

In this section, we design the experimental setup, and analyze the setting of the tradeoff parameter. To evaluate the performance of the proposed method, four hyperspectral data sets captured by three different sensors are used to evaluate the performance of our detection method, which are listed as follows.

HYDICE Data Set: This data set was acquired by the Hyperspectral digital imagery collection experiment (HYDICE) sensor over an urban area in CA, USA. A subimage of size 80×100 is widely used in many publications. This data set has 210 bands, but only 162 bands were left after removing the noisy bands. Among them, 21 pixels are considered as targets, which are cars and roofs.

San Diego Data Set: Another real HSI is also widely used in the related publications, which is collected by the AVIRIS sensor. The AVIRIS sensor collects spectral data in 224 bands spanning from 400 to 2500 nm. The scene is part of the San Diego airport area, CA, USA. This image has 224 spectral bands, and after removing the water vapor absorption and noisy bands, 189 bands remain. This data contains 100×100 pixels in each band. All the compared algorithms are used to detect three aircraft including 58 target pixels.

Cuprite Data Set: The Cuprite[1] data were also acquired by the AVIRIS sensor in the Cuprite mining district of NV, USA, in 1997. Here, a subset of Cuprite data containing 250×191 pixels and 188 bands (low-SNR and water absorption bands were removed) is utilized in the experiments. The buddingtonite is considered as the target that should be detected in this data.

Airport–Beach–Urban (ABU) Database: The ABU database was made available online.[2] The detailed acquisition procedure has been described in [44]. These sample images contain 100×100 or 150×150 pixels with the corresponding references. The airplane and urban scenes were captured by the AVIRIS sensor, and most of the beach scenes are also captured by the AVIRIS sensor. One of the beach scenes was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor.

To verify the effectiveness of the proposed method, three representative target detection methods, including CEM, hCEM, CSCR, and ACE, are utilized for comparison. These competing methods are either frequently cited in the literature or have leading performances on several real hyperspectral data sets. The proposed SRUN-AE and SRUN-VAE models are evaluated by the subsequent detection. The performance of the proposed method is evaluated by the most widely used metric, the receiver operating characteristic (ROC), and the area under the curve (AUC) metric [45]. Based on the reference map, the ROC curve gives the relationship between the true-positive rate and false positive rate by changing different thresholds on a detector's output.

B. Effect of Parameters

The dimensionality of the latent feature matrix is the number of hidden nodes in the deepest code layer, which determines the quality of the learned features. The depth of the network and the regularization parameter α are two other significant parameters for effective feature extraction. As mentioned earlier, α can penalize large L_s. Besides, the threshold τ

[1] http://aviris.jpl.nasa.gov/html/aviris.freedata.html
[2] http://xudongkang.weebly.com/

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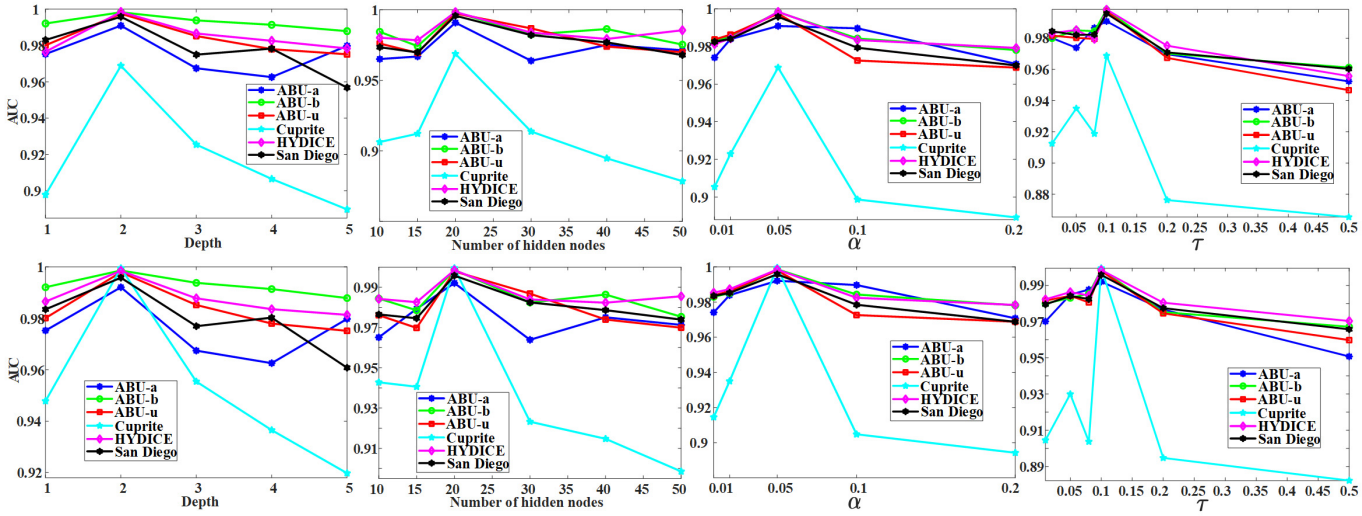


Fig. 5. Parameter sensitivity analysis for the parameters depth, number of hidden nodes, α, and τ. The first row and second row show the relationship between these four parameters and AUC values obtained by SRUN-AE and SRUN-VAE, respectively.

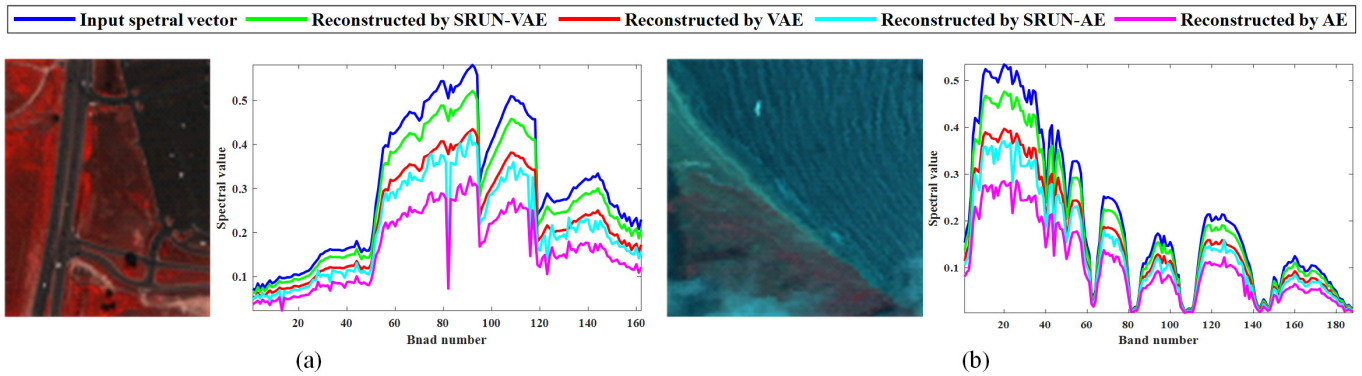


Fig. 6. Reconstructed target spectra operated by SRUN-VAE, SRUN-AE, VAE, and AE in the (a) HYDICE data and (b) ABU-beach3 data, respectively.

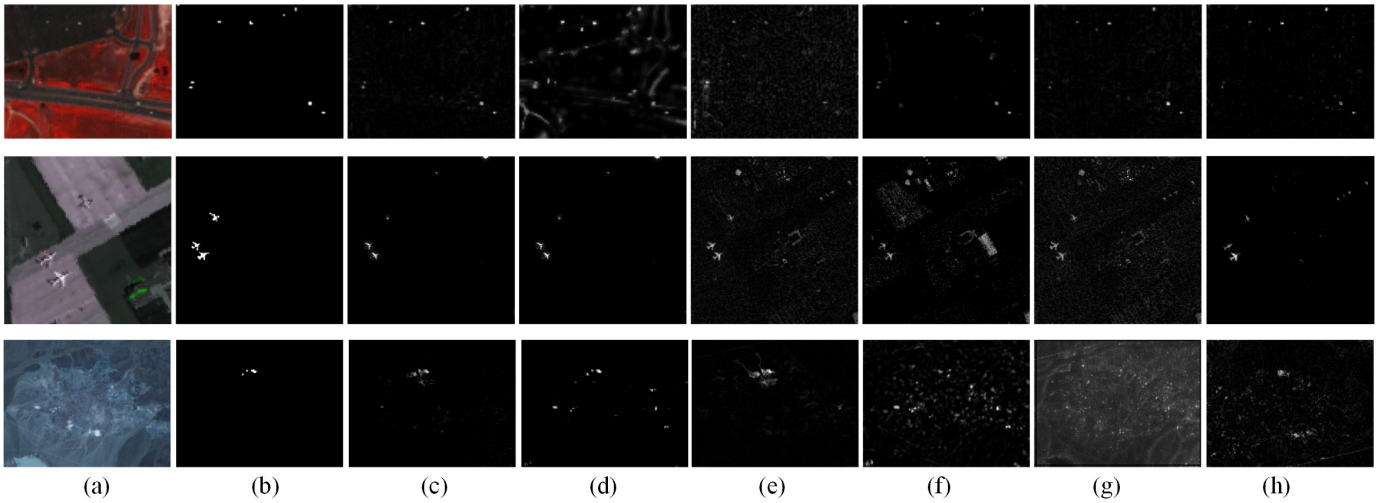


Fig. 7. Detection maps of the compared methods. (a) Color composites of the HYDICE, San Diego, and Cuprite images, (b) reference map, (c) SRUN-VAE, (d) SRUN-AE, (e) CEM, (f) hCEM, (g) CSCR, and (h) ACE.

in the feature selection procedure determines whether the target feature maps are sufficiently selected. If τ is determined correctly, the targets of different scales will be detected. In specific, Fig. 5 shows the effect of these parameters on the AUC values of target detection. Here, the parameters of both SRUN-VAE and SRUN-AE models are all analyzed.

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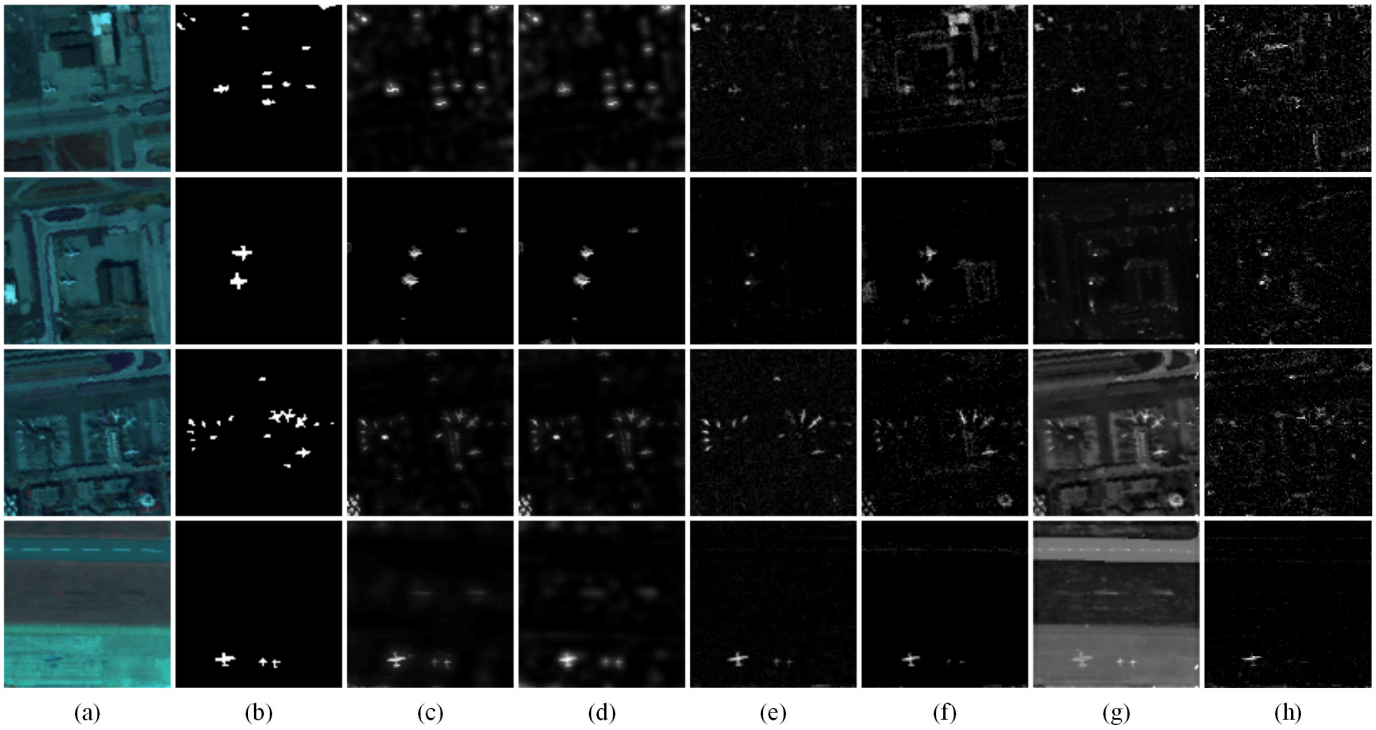


Fig. 8. Detection maps of the compared methods. (a) Color composites of the airport scene in ABU data set, (b) reference map, (c) SRUN-VAE, (d) SRUN-AE, (e) CEM, (f) hCEM, (g) CSCR, and (h) ACE.

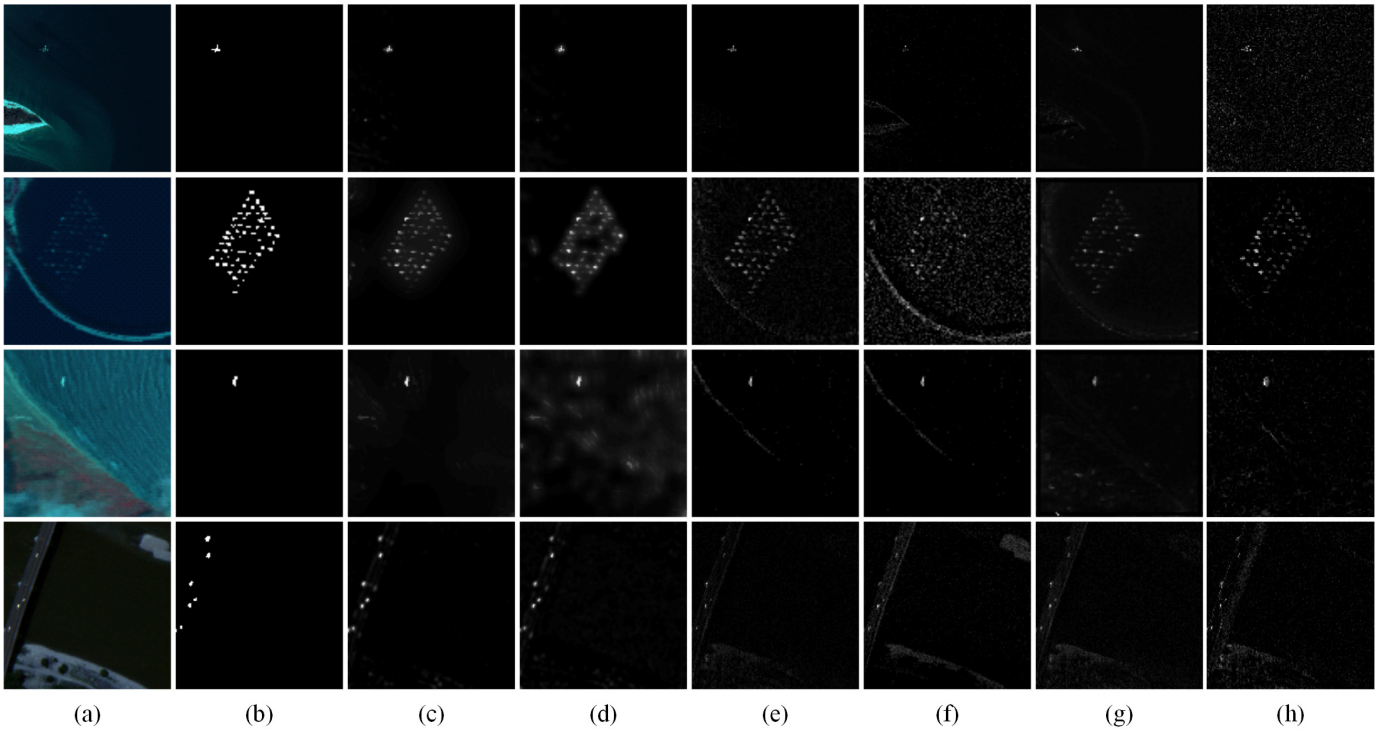


Fig. 9. Detection maps of the compared methods. (a) Color composites of the beach scene in ABU data set, (b) reference map, (c) SRUN-VAE, (d) SRUN-AE, (e) CEM, (f) hCEM, (g) CSCR, and (h) ACE.

The first row and second row show the relationship between four parameters and AUC values obtained by SRUN-AE and SRUN-VAE, respectively. It can be observed that, when the depth reaches 2, the number of hidden nodes is 20, α is 0.05, and τ is 0.1, and our proposed SRUN model can achieve best detection performance. Consequently, we fix the depth as 2,

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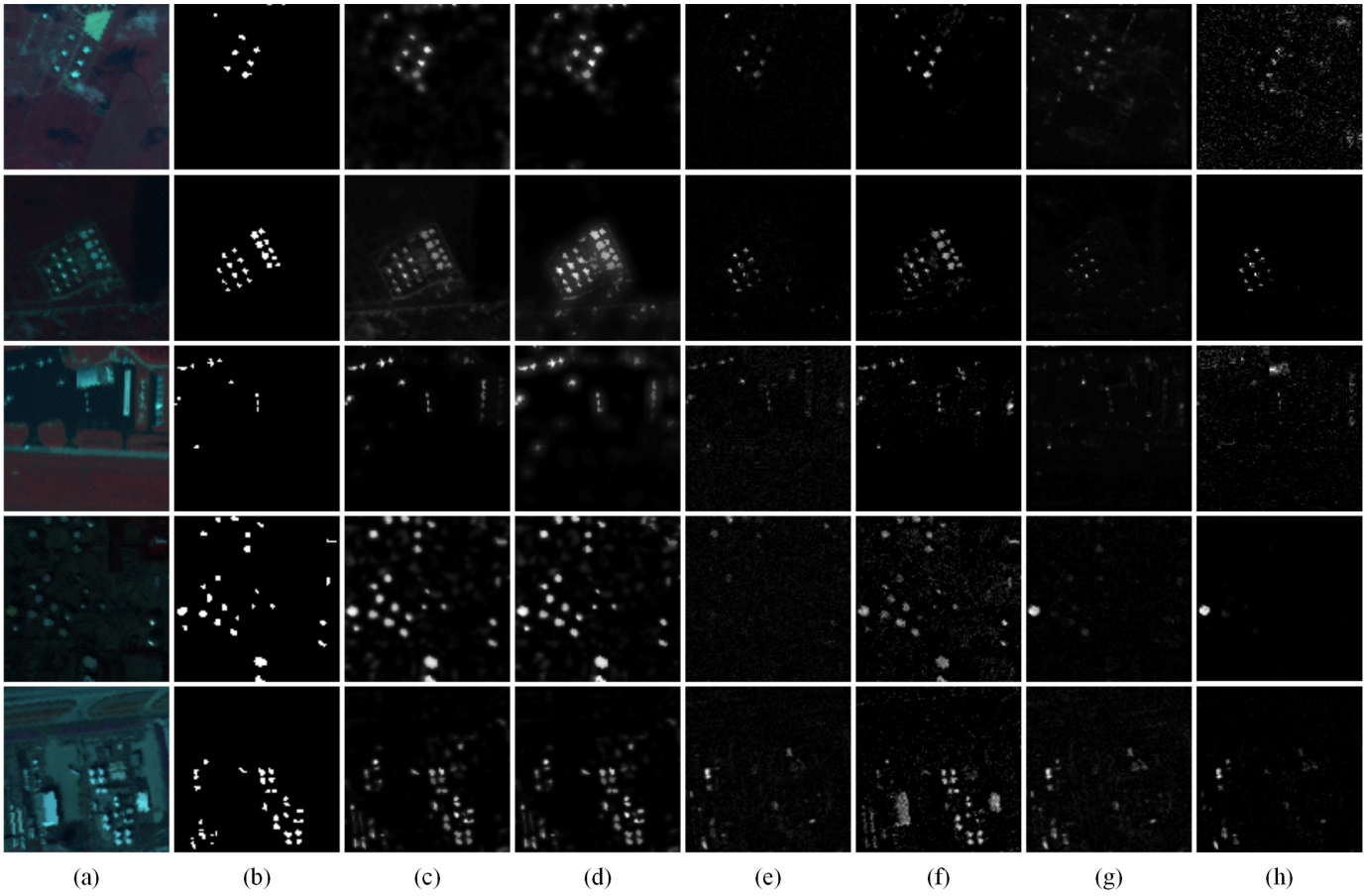


Fig. 10. Detection maps of the compared methods. (a) Color composites of the urban scene in ABU data set, (b) reference map, (c) SRUN-VAE, (d) SRUN-AE, (e) CEM, (f) hCEM, (g) CSCR, and (h) ACE.

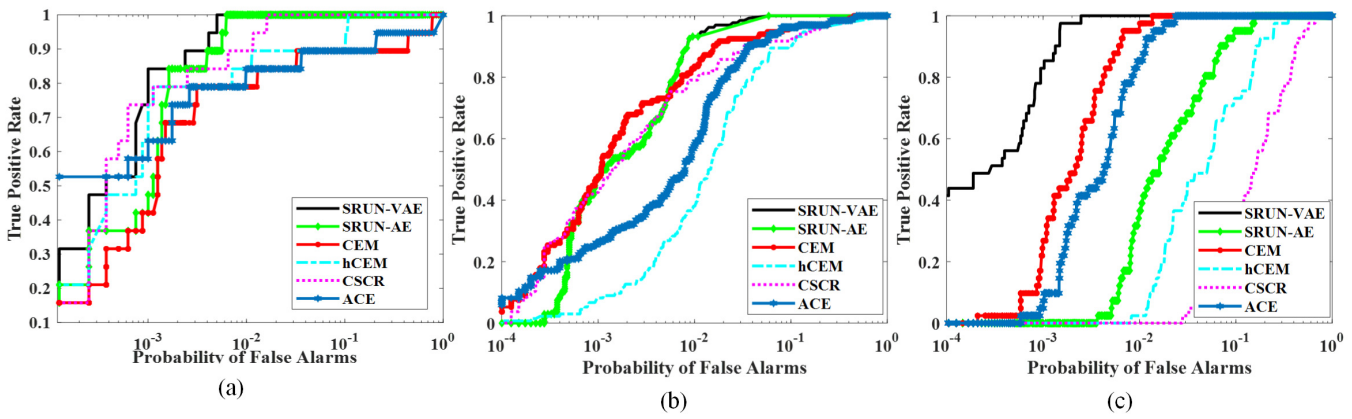


Fig. 11. ROC curves of the compared methods on the (a) HYDICE, (b) San Diego, and (c) Cuprite data sets.

the number of hidden nodes is 20, α is 0.05, and τ is 0.1 for all data sets.

C. Ablation Study

With respect to HSIs, we impose a spectral constraint on the original loss of AE and VAE to regularize spectral information. Fig. 6 demonstrates the reconstructed spectral vector of two target pixels operated by the proposed SRUN and the original VAE and AE models, respectively. Compared with the reconstruction by the original AE and VAE, the spectrum reconstructed by the proposed SRUN-AE and SRUN-VAE can significantly preserve spectral information of the input, which shows the extracted feature nodes retain enough information

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TABLE I
AUC VALUES FOR THE COMPARED METHODS ON DIFFERENT DATA SETS

Images	SRUN-VAE	SRUN-AE	CEM	hCEM	CSCR	ACE
HYDICE	**0.99846**	0.99841	0.93327	0.98745	0.99835	0.97391
San Diego	**0.99579**	**0.99579**	0.98389	0.94149	0.97957	0.97373
Cuprite	**0.99950**	0.96880	0.99834	0.92538	0.79981	0.99693
ABU-a1	**0.98854**	0.98689	0.79062	0.87304	0.87208	0.73657
ABU-a2	**0.99297**	0.99266	0.87500	0.95334	0.86261	0.78320
ABU-a3	**0.98748**	0.98546	0.90722	0.92045	0.89842	0.74684
ABU-a4	**0.99929**	0.99850	0.97522	0.89384	0.99264	0.86785
ABU-b1	**0.99989**	0.99986	0.99320	0.88695	0.98596	0.85643
ABU-b2	**0.99519**	0.99386	0.96479	0.82485	0.97446	0.96824
ABU-b3	**0.99997**	0.99996	0.99965	0.95197	0.97309	0.99963
ABU-b4	**0.99926**	0.99925	0.88818	0.85401	0.90376	0.79915
ABU-u1	**0.99903**	0.99889	0.98157	0.95270	0.99277	0.72190
ABU-u2	**0.99919**	0.99883	0.74943	0.99051	0.80675	0.72933
ABU-u3	**0.99854**	0.99697	0.92401	0.98535	0.99519	0.94320
ABU-u4	**0.99730**	0.99723	0.80839	0.98753	0.90068	0.76476
ABU-u5	**0.99545**	0.99534	0.76561	0.94784	0.85072	0.86924

from the input, and thus, can better represent the spectra of targets. Besides, as we can see, the reconstructed spectra of targets by the proposed SRUN-VAE are more similar to the original ones, and the reconstruction effect of VAE is better than that of SRUN-AE. These phenomena show that Gaussianity is more beneficial for recovering spectral vectors. In summary, both spectral regularization loss and Gaussianity are useful for reconstructing HSIs.

As shown in Fig. 4, the response of background is much larger than that of the target in some feature maps, which is not conducive to target detection. After feature selection, the target feature maps (highlighted with yellow in Fig. 4) can present more distinctive patterns as compared with that of the background. The weighting process is to combine the complementary information of the selected feature maps, and it is more efficient to conduct target detection on the weighted feature map than a feature matrix with C dimensions.

D. Detection Results

The detection results in visual for four data sets by the different sensors are shown in Figs. 7–10. As expected, the proposed SRUN-based detection method causes the smaller visual difference between the reference and detection map compared with CEM, hCEM, CSCR, and ACE. More specifically, the features extracted by the proposed SRUN-VAE model have the best effect on target detection in visual. Besides, our proposed method can detect the targets of different sizes. The CEM and ACE methods area typical target detection methods, which cannot perform well for the targets containing structural information. The hCEM and CSCR methods are recently proposed target detection methods, which usually have stable detection performances, while they may enhance background pixels for some sample images, such as Figs. 7 and 9. In comparison with these methods, the major advantage of the proposed method is its robust detection performances for different targets with different scales in all scenes. Furthermore, the reported AUC values in Table I on the HYDICE, San Diego, Cuprite, and ABU data sets comply with this observation. Therefore, we can conclude that the proposed method can achieve superior performances for all data sets. Besides, the proposed SRUN model based on VAE is better than AE in both subjective and objective evaluation.

With the optimal parameters, Fig. 11 shows the ROC performance of different detectors using three experimental data sets. From the results, it is obvious that the proposed SRUN-VAE is always superior to CEM, hCEM, CSCR, and ACE, because the curves of the detection results remain over those compared methods. This result confirms that our method has promising results for these three scenes. Although the CEM detector achieves better performance than the proposed SRUN-AE method with respect to the Cuprite data set, the proposed SRUN-AE method is second only to SRUN-VAE for all the other data sets. This phenomenon shows that the proposed detectors based on SRUN-VAE and SRUN-AE have better generalization ability. In general, the detection results are consistent with the AUC values reported in Table I.

V. CONCLUSION

In this article, VAE was studied for HSI unsupervised feature extraction for the first time. Specifically, spectral regularization is imposed into the unsupervised networks, such as VAE and AE, to extract intrinsic properties underlying HSIs. In this way, the characteristics of a high-dimensional HSI can be well represented and reduced by the proposed SRUN model. Besides, an effective and simple feature selection is presented, which depends on the spectral difference between the prior know spectrum and spectra of other pixels in input. In addition, the selected features are weighted adaptively to obtain a final detection map. The experiments on several real data sets show that the proposed method is effective for target detection and superior to the existing methods in both objective and subjective evaluation. Furthermore, the SRUN based on VAE can achieve superior performance than the SRUN based on AE in terms of spectral consistency and detection accuracy.

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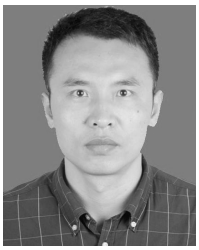
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